leads to lower capacity closer to the root node, a trend clearly visible in Figure S.4, i.e. lower depth (e.g. depth 2) has lower capacity than the higher depth (e.g. depth 5). The positive correlation between the capacity and the accuracies is clearly shown in Figure S.5. Finally, we also find that the attention map for this case (Figure S.6) is qualitatively the same as the regular prompt case shown in the main text.

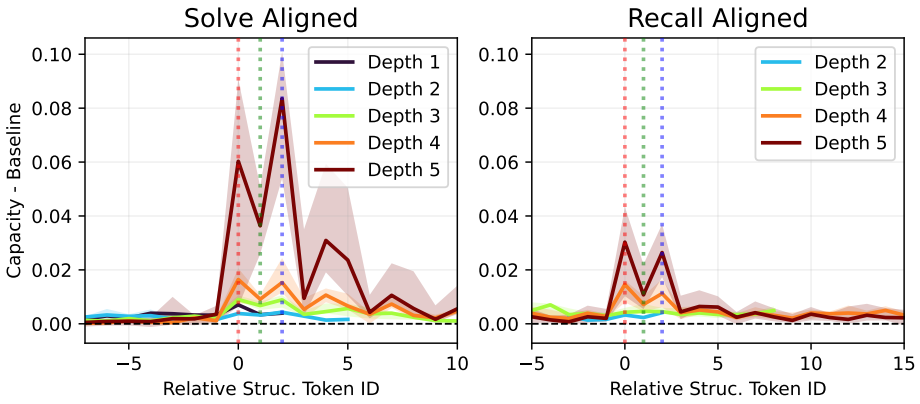

Figure S.4: The masked logic case. Capacity values for each node grouped by the logic tree depth. Left: the node capacity traces aligned at the solve phases of the corresponding nodes. Right: the node capacity traces aligned at the recall phases of the parent nodes (the main focus of this figure).

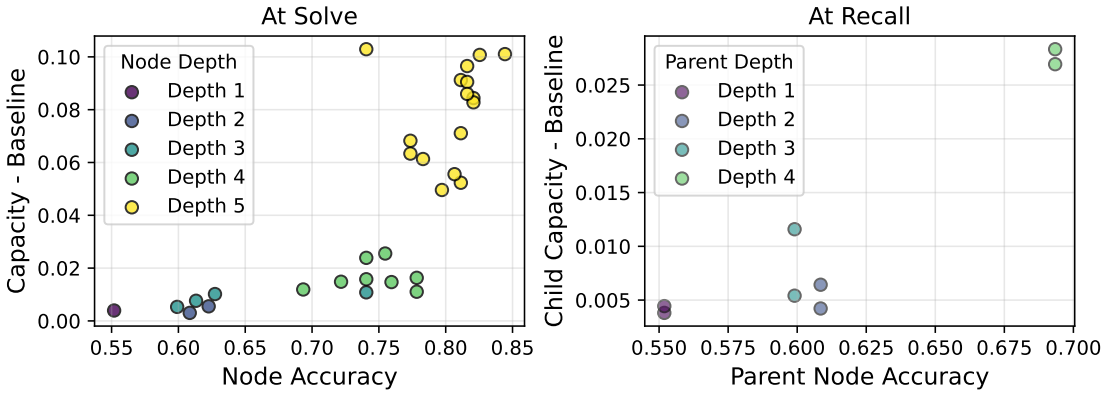

Figure S.5: The masked logic case. Left: node capacity at its solve phase vs. corresponding node accuracy. Right: child node capacity at its recall phase vs. its parent node accuracy.

C Hyperplane Direction Analysis

We computed the max-margin hyperplanes at each token of interest based on each node's ground truth labels, and analyzed some pairwise cosine similarities. First, we check if, for all nodes, the manifolds separate along the same direction when they are being solved. It turns out that at the "*Logic:" token of the solve phase, only the leaf nodes have highly conserved directions, but for the rest, the directions are mostly orthogonal Figure S.7-**top-left**. This reveals that, when a node is being actively solved, each node takes a new orthogonal direction. However, at the "*Result:" token, all nodes point toward the same direction, presumably the direction that encodes True or False Figure S.7-**bottom-left**.

We also check if the recall direction is conserved across nodes Figure S.7-**middle-and-right**. As noted in the main text, we see that the recall directions are highly conserved across all left child nodes and also for all right child nodes Figure S.7-**right**.. Regardless of where in the tree the LM is solving, it treats the left and right child nodes of the node that is being solved equally. This indicates that the LM applies the same local rule to compute all subproblems. These two directions (one for the left child and another for the right child) are slightly anti-correlated.

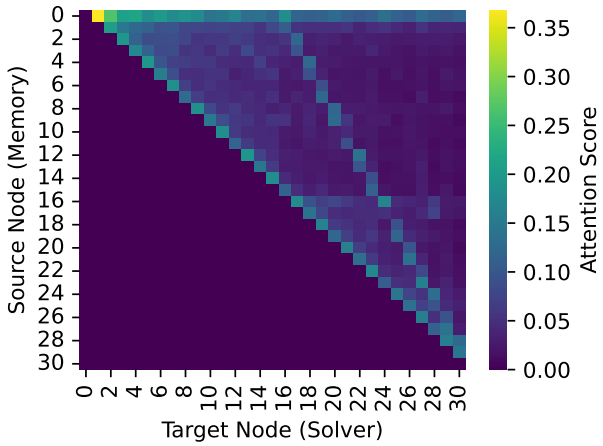

Figure S.6: Attention score heatmap between graph nodes for the masked logic case. The x-axis represents the target node currently being solved; the y-axis represents the source nodes in the context. The diagonal structure reflects the model attending to recent context, while off-diagonal points represent the retrieval of specific dependencies.

Finally, we see if, for a given node, the directions when it is solved and being recalled are conserved. It turns out that the solving and recalling directions are mostly orthogonal Figure S.8.

D Quantification of Attention Scores

To investigate whether the model retrieves information from previous reasoning steps, we analyze the attention patterns during the "Solve" phase of each parent node. We define the attention score $A_{T \to S}$ from a target node T (the parent) to a source node S (the child) as follows.

Let $\mathbf{M}^{(l)} \in \mathbb{R}^{H \times N \times N}$ be the multi-head attention matrix at layer l, where H is the number of heads and N is the sequence length. We identify the set of token indices corresponding to the source node's solution phase, $\mathcal{I}_S = \{t \mid \text{token}_t \in \text{Result String of } S\}$, and the set of token indices corresponding to the target node's computation phase, $\mathcal{I}_T = \{t \mid \text{token}_t \in \text{Logic String of } T\}$.

We compute the maximum attention score across all heads and all token pairs within the defined windows:

$$A_{T \to S} = \max_{l \in L_{\text{ROI}}} \left[\max_{h \in \{1 \ldots H\}} \left(\max_{i \in \mathcal{I}_T, j \in \mathcal{I}_S} \mathbf{M}^{(l)}_{h,i,j} \right) \right] \tag{1}$$

where L_{ROI} represents the layers of interest (e.g., Layer 20). The maximum aggregation is chosen over the mean to detect sparse retrieval mechanisms, as information transfer in transformers is often performed by specialized induction heads rather than distributed uniformly across all heads.

For the correlation analysis shown in Figure 6b, we align this attention score $A_{T \to S}$ with the manifold capacity of the source node S, $\alpha^{(S)}$, measured using the embeddings at the last token of the target phase $\mathcal{I}_T$. This allows us to directly compare the strength of retrieval with the resulting linear separability of the retrieved concept.

E Additional Capacity and Probe Traces

Please see Figures S.9 and S.10.

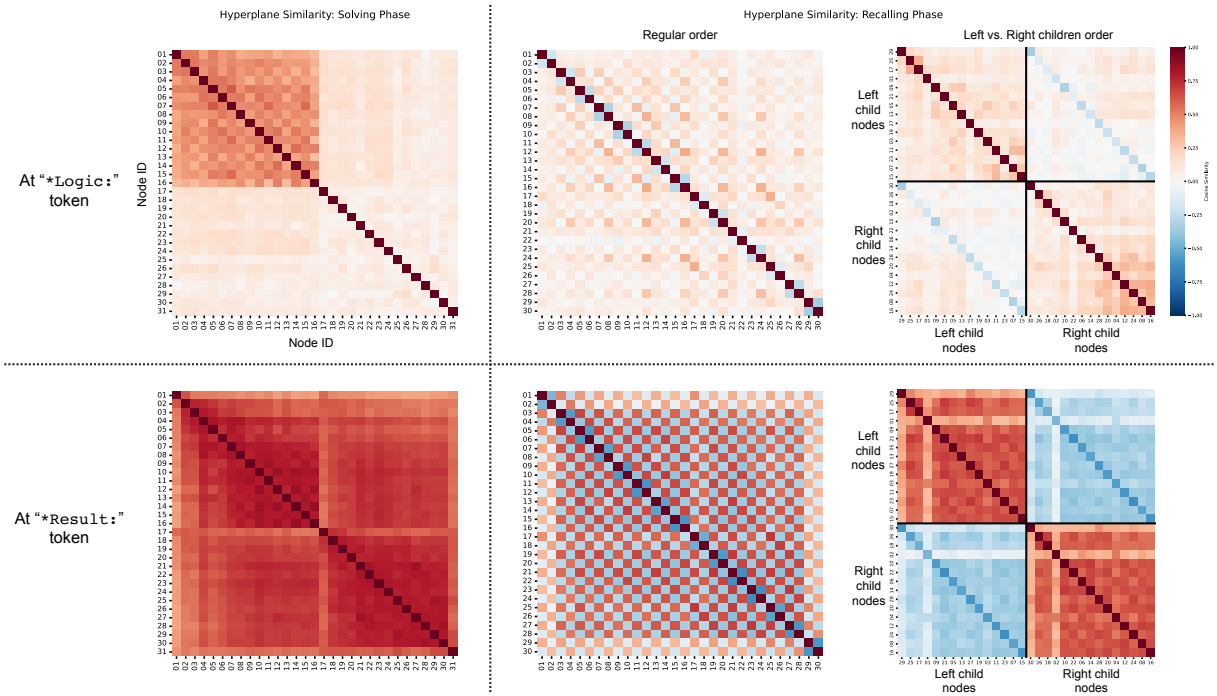

Figure S.7: **Cosine Similarity of Hyperplane Directions between Nodes when They are Solved or Recalled.** Cosine similarity between the max-margin hyperplanes computed for nodes' ground truth answers at the "*Logic:" tokens (for the **top** row; "*Result:" tokens for the **bottom** row) in its solve phase (for the **left** column; recalling phase for the **middle** column). For the recall phase similarity, we also show the same plot but with the left and right child nodes grouped, revealing the block structure (the **right** column).